

CHOSHI, Japan

WMO: 476480

Lat: 35.7397N

Lon: 140.8572E

Elev: 28

StdP: 100.99

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	1.4	-10.2	1.6	4.1	-8.8	1.8	4.2	18.2	7.2	15.5	7.9	5.6	290	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	30.2	26.0	29.4	25.7	28.5	25.3	26.7	28.9	26.2	28.3	25.7	27.7	6.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.5	27.8	25.6	20.8	27.5	25.1	20.2	27.2	83.8	29.0	81.5	28.4	79.5	27.9	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	12.9	11.5	-1.4	32.4	1.0	1.1	-2.1	33.2	-2.7	33.9	-3.2	34.5	-4.0	35.3		
(5)			-3.8	27.4	1.0	0.6	-4.5	27.8	-5.1	28.2	-5.7	28.5	-6.4	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	6.4	6.7	9.6	13.8	17.6	20.3	23.9	25.9	23.5	19.3	14.1	9.1
	DBStd	7.08	2.36	2.58	3.03	2.81	2.40	2.16	2.34	2.05	2.43	2.34	2.65	2.79
	HDD10.0	317	116	98	46	4	0	0	0	0	0	0	2	51
	HDD18.3	1583	371	326	272	137	42	7	1	0	1	16	127	285
	CDD10.0	2474	3	6	33	119	235	310	431	494	406	288	126	23
	CDD18.3	699	0	0	0	1	18	67	173	236	157	46	1	0
	CDH23.3	3831	0	0	0	0	10	101	951	1903	804	62	0	0
	CDH26.7	716	0	0	0	0	0	3	163	428	121	2	0	0
(14) Wind	WSAvg	5.7	5.7	6.0	6.4	6.2	5.6	4.9	5.2	5.1	5.8	6.2	5.6	5.5
(15) Precipitation	PrecAvg	1660	90	94	144	126	142	166	107	116	205	244	135	85
	PrecMax	2356	273	233	369	247	359	329	312	352	468	708	394	296
	PrecMin	1068	7	16	42	36	28	35	12	0	17	55	14	4
	PrecStd	257	60	52	59	47	66	67	67	87	101	139	81	54
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	17.0	18.7	21.5	24.5	27.0	30.7	31.7	30.3	26.6	21.7	19.4
		MCWB	12.9	14.3	14.4	17.3	19.6	23.2	25.8	26.4	25.6	23.2	18.3	16.9
	2%	DB	13.8	14.7	17.2	20.0	23.2	25.6	29.5	30.7	29.2	24.9	20.5	16.9
		MCWB	10.5	12.0	14.1	16.5	19.0	22.2	25.5	26.2	25.5	22.1	17.6	13.9
	5%	DB	12.1	12.8	16.0	19.0	22.2	24.5	28.5	29.9	28.2	23.7	19.5	15.1
		MCWB	8.8	9.7	13.2	15.8	18.5	21.5	25.1	26.0	25.1	20.9	16.6	11.9
	10%	DB	10.6	11.2	14.7	18.0	21.2	23.5	27.4	29.1	27.0	22.7	18.5	13.8
		MCWB	7.3	8.0	11.9	15.1	18.1	21.1	24.7	25.7	24.6	19.9	15.6	10.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	15.1	16.2	18.9	21.2	24.0	26.6	27.6	26.7	24.2	19.8	17.7
		MCDB	15.3	16.4	17.2	19.8	22.6	25.8	29.2	29.7	28.6	25.7	20.8	18.9
	2%	WB	11.5	12.7	14.9	17.7	20.3	23.1	26.0	27.0	26.1	23.0	18.5	14.6
		MCDB	13.1	14.2	16.6	19.3	21.9	24.6	28.4	29.3	28.0	24.5	19.8	16.3
	5%	WB	9.4	10.4	13.8	16.7	19.6	22.4	25.5	26.5	25.6	21.7	17.4	12.6
		MCDB	11.4	12.2	15.7	18.3	21.2	23.7	27.6	28.7	27.4	23.1	18.8	14.5
	10%	WB	7.7	8.6	12.2	15.6	18.8	21.7	25.0	26.1	25.0	20.6	16.3	11.2
		MCDB	10.2	10.7	14.3	17.4	20.5	23.0	26.9	28.2	26.7	22.1	17.9	13.3
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.2	5.8	5.5	5.1	4.5	4.8	4.7	4.0	4.2	5.6	6.4
		MCDBR	8.4	8.4	7.2	6.7	6.4	5.7	6.4	5.7	5.4	5.4	6.5	7.7
	5% WB	MCWBR	7.0	7.4	6.5	4.8	3.7	3.1	3.0	2.4	2.8	4.2	5.5	6.5
		MCWBR	7.7	7.7	6.7	5.9	5.8	5.1	5.9	5.5	5.1	4.9	5.9	7.0
(40) Clear-Sky Solar Irradiance	taub		0.382	0.434	0.501	0.547	0.557	0.576	0.573	0.547	0.493	0.473	0.427	0.385
		taud	2.240	2.091	1.933	1.870	1.903	1.928	1.984	2.051	2.174	2.139	2.184	2.246
	Ebn at Noon		792	787	765	754	752	737	735	746	767	739	733	759
		Edh at Noon	107	139	178	199	195	190	179	164	139	132	113	101
(44) All-Sky Solar Radiation	RadAvg		2.59	3.05	3.93	4.67	5.08	4.60	5.10	5.19	3.96	2.96	2.37	2.26
	RadStd		0.18	0.34	0.28	0.47	0.50	0.42	0.83	0.49	0.33	0.28	0.21	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page